

Assignment 8: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

NOTE: For this assignment, you can skip tests for any of the assumptions required to run the models.

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

library(lubridate)
library(agricolae)
library(here)

## here() starts at /home/guest/EDE_Spring2026

here()

## [1] "/home/guest/EDE_Spring2026"

options(scipen = 4)

NTL.chemphys.raw <- read.csv(
  here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
  stringsAsFactors = TRUE
)

# convert sampleddate from factor to character, then to Date
NTL.chemphys.raw <- NTL.chemphys.raw %>%
  mutate(sampledate = mdy(as.character(sampledate)))

#2
my_theme <- theme_classic(base_size = 14) +
  theme(
    axis.text = element_text(color = "black"),
    legend.position = "top"
  )

theme_set(my_theme)

```

Simple regression

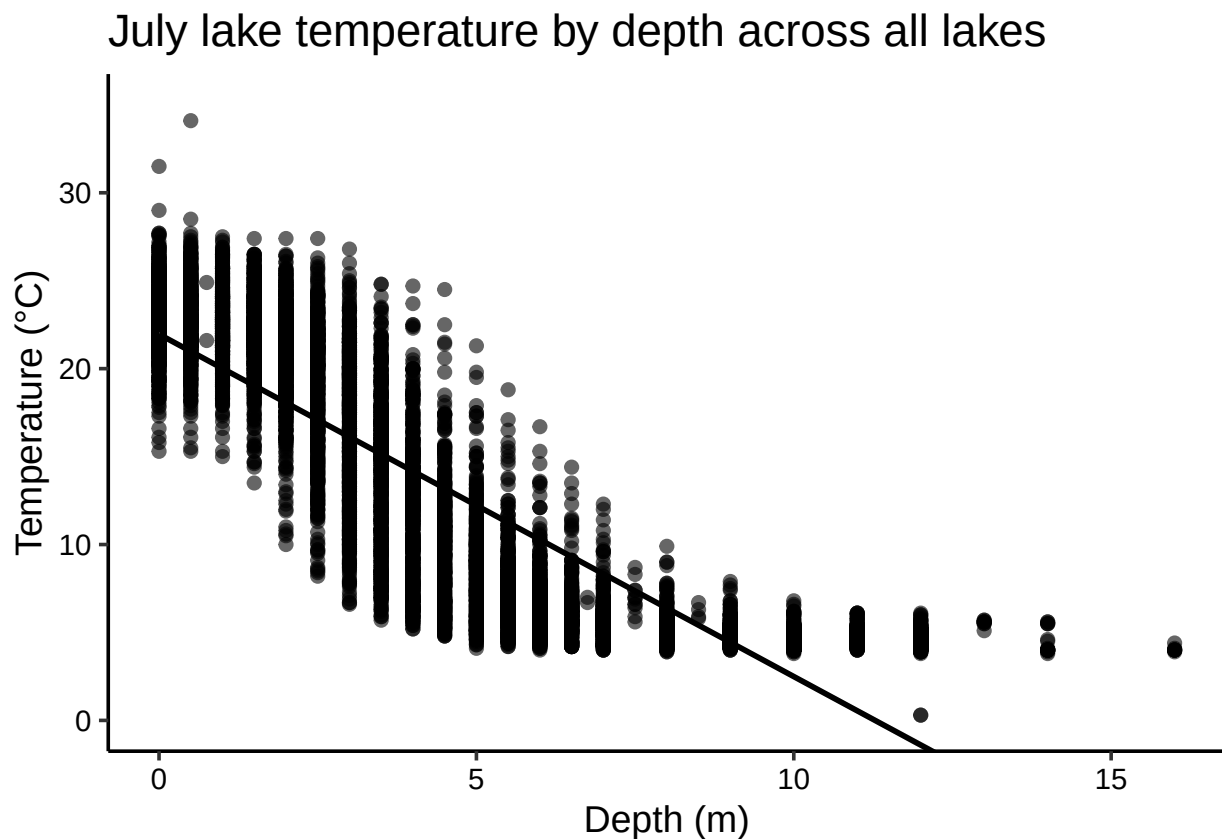
Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: There is no relationship between July mean lake temperature and depth across lakes; the slope of temperature on depth is 0.
Ha: July mean lake temperature changes with depth across lakes; the slope of temperature on depth is not 0.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
July.temp <- NTL.chemphys.raw %>%
  filter(month(sampledate) == 7) %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  drop_na()

#5
ggplot(July.temp, aes(x = depth, y = temperature_C)) +
  geom_point(alpha = 0.6, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  coord_cartesian(ylim = c(0, 35)) +
  labs(
    title = "July lake temperature by depth across all lakes",
    x = "Depth (m)",
    y = "Temperature (°C)"
  )
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The figure suggests that temperature generally decreases as depth increases, so deeper water is colder in July. The points are more spread out near the surface, while deeper observations

are more consistently cold. The trend does not look perfectly linear, but a negative linear relationship seems like a reasonable summary of the overall pattern.

7. Perform a linear regression to test the relationship and display the results.

```
#7
temp.depth.lm <- lm(temperature_C ~ depth, data = July.temp)
summary(temp.depth.lm)

##
## Call:
## lm(formula = temperature_C ~ depth, data = July.temp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16

cor.test(July.temp$temperature_C, July.temp$depth)

##
## Pearson's product-moment correlation
##
## data:  July.temp$temperature_C and July.temp$depth
## t = -165.83, df = 9726, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.8646036 -0.8542169
## sample estimates:
##          cor
## -0.8594989
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The regression shows that July temperature decreases significantly with depth. Depth explains about XX% of the variation in temperature. This result is based on $df = X$ and Y and is statistically significant ($p < 0.001$). The model predicts that for every 1 m increase in depth, temperature decreases by about X.XX °C on average.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
temp.full.lm <- lm(temperature_C ~ year4 + daynum + depth, data = July.temp)

temp.aic <- step(temp.full.lm, direction = "both", trace = FALSE)
temp.aic
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = July.temp)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##   -8.57556      0.01134      0.03978     -1.94644
```

```
#10
summary(temp.aic)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = July.temp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994   0.32044
## year4        0.011345   0.004299   2.639   0.00833 **
## daynum       0.039780   0.004317   9.215  < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC procedure suggests keeping year4, daynum, and depth in the final model. This model explains 74.1% of the observed variation in July temperature. Compared with the model using only depth (73.9%), this is only a slight improvement, but it still provides a somewhat better fit.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
lake.temp.aov <- aov(temperature_C ~ lakename, data = July.temp)
summary(lake.temp.aov)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lake.temp.lm <- lm(temperature_C ~ lakename, data = July.temp)
summary(lake.temp.lm)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = July.temp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake        -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake   -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

```
anova.tab <- summary(lake.temp.aov)[[1]]
anova.p <- anova.tab[1, "Pr(>F)"]
anova.df1 <- anova.tab[1, "Df"]
anova.df2 <- anova.tab[2, "Df"]
anova.F <- anova.tab[1, "F value"]
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: There is a significant difference in mean July temperature among the lakes (ANOVA, $F(8, 9719) = 50.0$, $p < 0.001$). This suggests that at least some lakes differ from one another in their average July temperature.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

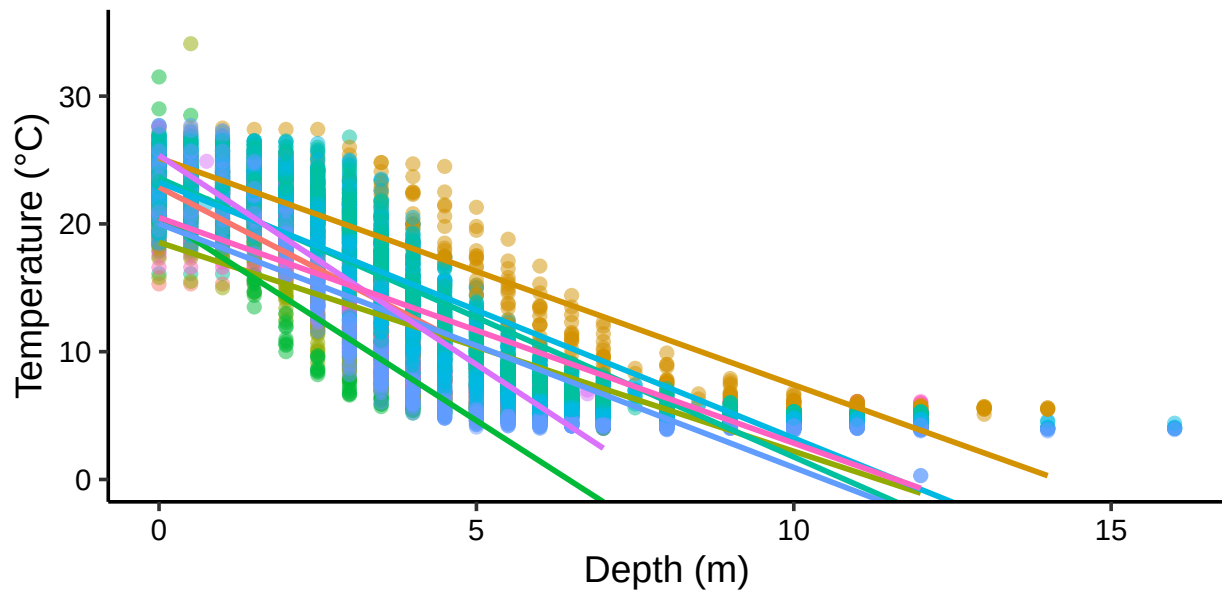
```
#14.
temp.depth.lake.plot <- ggplot(July.temp, aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point(alpha = 0.5, size = 2) +
  geom_smooth(method = "lm", se = FALSE) +
  coord_cartesian(ylim = c(0, 35)) +
  labs(
    title = "July temperature by depth for each lake",
    x = "Depth (m)",
    y = "Temperature (°C)",
    color = "Lake"
  )

print(temp.depth.lake.plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

July temperature by depth for each lake

Central Long Lake East Long Lake Paul Lake Tuesday Lake West Long Lake
Crampton Lake Hummingbird Lake Peter Lake Ward Lake



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
lake.tukey <- TukeyHSD(lake.temp.aov)
lake.tukey
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = July.temp)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999


```
## Tuesday Lake-Crampton Lake      -4.2826611 -5.6895065 -2.8758157 0.0000000
## Ward Lake-Crampton Lake          -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake     -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake  0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake         3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake        3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake      0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake         4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake    1.3109897  0.2885003  2.3334791 0.0022805
## Paul Lake-Hummingbird Lake       3.0409798  0.8765299  5.2054296 0.0004495
## Peter Lake-Hummingbird Lake      2.5429846  0.3818755  4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake    0.2959499 -1.9019508  2.4938505 0.9999752
## Ward Lake-Hummingbird Lake       3.6853448  0.6889874  6.6817022 0.0043297
## West Long Lake-Hummingbird Lake  0.8053791 -1.4299320  3.0406903 0.9717297
## Peter Lake-Paul Lake             -0.4979952 -1.1120620  0.1160717 0.2241586
## Tuesday Lake-Paul Lake          -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake             0.6443651 -1.5200848  2.8088149 0.9916978
## West Long Lake-Paul Lake        -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake         -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake            1.1423602 -1.0187489  3.3034693 0.7827037
## West Long Lake-Peter Lake       -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake          3.3893950  1.1914943  5.5872956 0.0000609
## West Long Lake-Tuesday Lake     0.5094292 -0.4121051  1.4309636 0.7374387
## West Long Lake-Ward Lake        -2.8799657 -5.1152769 -0.6446546 0.0021080
```

```
lake.groups <- HSD.test(lake.temp.aov, "lakename", group = TRUE)
lake.groups
```

```
## $statistics
##      MSerror  Df      Mean      CV
##    54.1016 9719 12.72087 57.82135
##
## $parameters
##      test name.t ntr StudentizedRange alpha
##    Tukey lakename  9          4.387504  0.05
##
## $means
##               temperature_C      std      r      se Min  Max   Q25   Q50
## Central Long Lake      17.66641 4.196292  128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake          15.35189 7.244773  318 0.4124692 5.0 27.5  7.525 16.90
## East Long Lake         10.26767 6.766804  968 0.2364108 4.2 34.1  4.975  6.50
## Hummingbird Lake       10.77328 7.017845  116 0.6829298 4.0 31.5  5.200  7.00
## Paul Lake              13.81426 7.296928 2660 0.1426147 4.7 27.7  6.500 12.40
## Peter Lake              13.31626 7.669758 2872 0.1372501 4.0 27.0  5.600 11.40
## Tuesday Lake           11.06923 7.698687 1524 0.1884137 0.3 27.7  4.400  6.80
## Ward Lake              14.45862 7.409079  116 0.6829298 5.7 27.6  7.200 12.55
## West Long Lake         11.57865 6.980789 1026 0.2296314 4.0 25.7  5.400  8.00
##
##               Q75
## Central Long Lake 21.000
## Crampton Lake    22.300
## East Long Lake    15.925
## Hummingbird Lake 15.625
## Paul Lake         21.400
## Peter Lake        21.500
```

```
## Tuesday Lake      19.400
## Ward Lake         23.200
## West Long Lake    18.800
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake    17.66641      a
## Crampton Lake        15.35189     ab
## Ward Lake            14.45862     bc
## Paul Lake            13.81426      c
## Peter Lake           13.31626      c
## West Long Lake       11.57865      d
## Tuesday Lake         11.06923     de
## Hummingbird Lake     10.77328     de
## East Long Lake        10.26767      e
##
## attr("class")
## [1] "group"
```

```
# extract pairwise comparisons involving Peter
peter.comp <- as.data.frame(lake.tukey$lakename) %>%
  rownames_to_column("comparison") %>%
  filter(str_detect(comparison, "Peter"))

peter.same <- peter.comp %>%
  filter(`p adj` >= 0.05) %>%
  mutate(other_lake = case_when(
    str_starts(comparison, "Peter-") ~ str_remove(comparison, "Peter-"),
    str_ends(comparison, "-Peter") ~ str_remove(comparison, "-Peter"),
    TRUE ~ comparison
  )) %>%
  pull(other_lake)

peter.diff <- peter.comp %>%
  filter(`p adj` < 0.05) %>%
  mutate(other_lake = case_when(
    str_starts(comparison, "Peter-") ~ str_remove(comparison, "Peter-"),
    str_ends(comparison, "-Peter") ~ str_remove(comparison, "-Peter"),
    TRUE ~ comparison
  )) %>%
  pull(other_lake)
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: According to the Tukey test, the lakes that are not significantly different from Peter Lake are Paul Lake and Ward Lake. The lakes that are significantly different from Peter Lake are Central Long Lake, Crampton Lake, East Long Lake, Hummingbird Lake, Tuesday Lake, and West Long Lake. Based on the full Tukey output and grouping results, East Long Lake appears to be statistically distinct from all other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: A two-sample t-test would make sense here, since we would be comparing the mean temperature between just two lakes.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
Crampton.Ward <- July.temp %>%  
  filter(lakename %in% c("Crampton Lake", "Ward Lake")) %>%  
  droplevels()
```

```
Crampton.Ward %>%  
  count(lakename)
```

```
##      lakename    n  
## 1 Crampton Lake 318  
## 2      Ward Lake 116
```

```
cw.t.test <- t.test(temperature_C ~ lakename, data = Crampton.Ward)  
cw.t.test
```

```
##  
## Welch Two Sample t-test  
##  
## data:  temperature_C by lakename  
## t = 1.1181, df = 200.37, p-value = 0.2649  
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0  
## 95 percent confidence interval:  
## -0.6821129  2.4686451  
## sample estimates:  
## mean in group Crampton Lake      mean in group Ward Lake  
##                15.35189                14.45862
```

```
Crampton.Ward %>%  
  group_by(lakename) %>%  
  summarise(  
    mean_temp = mean(temperature_C),  
    sd_temp = sd(temperature_C),  
    n = n(),  
    .groups = "drop"  
  )
```

```
## # A tibble: 2 x 4  
##   lakename    mean_temp sd_temp    n  
##   <fct>      <dbl>    <dbl> <int>  
## 1 Crampton Lake    15.4     7.24   318  
## 2 Ward Lake       14.5     7.41   116
```

Answer: The two-sample t-test shows that the mean July temperatures of Crampton Lake and Ward Lake are not significantly different ($p = 0.265$). Based on this result, we do not reject the null hypothesis that the two lakes have the same mean temperature in July. This result is consistent with the Tukey HSD results from part 16. In the Tukey grouping output, Crampton Lake (group “ab”) and Ward Lake (group “bc”) share the letter “b”, indicating that their mean temperatures are not significantly different. Therefore, the t-test confirms the pattern already suggested by the ANOVA and Tukey analysis.